Tardive Dyskinesia Needs Assessment
Valbenazine (Ingrezza®) Overview

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Instructions:

- **Topic:** Tardive Dyskinesia (TD)
- **Supporter(s):** Neurocrine
- **Drugs:** valbenazine
- **Tactic:**
    - Topic for gap 1: Diagnosis and recognizing disease burden (impact of daily living, mental health etc)
    - Topic for gap 2: TD Clinical Data
    - Topic for gap 3: Addressing the disease burden with a multidisciplinary care team

- Please identify the gaps/challenges and root causes for the topics above and fill in the table.
- Please provide references for gaps and root causes as best you can.
- **Please write a few paragraphs (2-3) on the funder drug, valbenazine. The paragraphs should discuss the approved indication, important clinical data/data that may differentiate it from other therapies for TD, and the most recent/relevant data, which may come from news releases, pubmed, or data presented at a conference.**
- Please also mention the other drug(s) available for TD that would compete with this agent e.g. same drug class or similar indication etc. You do not need to include any data for this, just identify the other drug(s) available for TD.

- **Example:**
- **Pancreatic Cancer**

Gap #1: There remain misconceptions on the clinical impact of dose modifications of chemotherapy.			
Root Causes	**Learning Objectives**		
Studies have suggested that decreased relative dose intensity is associated with a poor prognosis, however, total dose intensity ≥ 62.5% has been shown to provide efficacy with better safety (Kobayashi 2022).	Describe potential clinical implications of dose modification strategies in pancreatic cancer.		
Gap #2: Clinicians lack standardized guidance on dose modifications for chemotherapy regimens.			
Root Causes	**Learning Objectives**		
Studies lack a unified pattern of dose modification and there is high diversity between 'planned' dose modifications and 'actually administered' dose (Jung 2023).	Summarize dose modification strategies to mitigate adverse events and improve time on chemotherapy in patients with pancreatic cancer		
Gap #3: Treatment-related gastrointestinal toxicities are extremely frequent, affect tolerability, and are the most bothersome of all AEs, but are often underreported.			
Root Causes	**Learning Objectives**		
First-line therapies have become more intensive with increased toxicity frequency and the burden on patient quality of life is not well-defined (Aprile 2015). Underreporting of adverse events may be attributed to differences in assessment techniques and scales. In a study of clinician-patient communication, instances of problematic understanding for patients and caregivers mainly occurred during discussion of diagnosis and treatment (Consolandi 2024).	Implement evidence-based approaches to proactively manage, monitor, recognize, and mitigate chemo-induced adverse events, with a focus on gastrointestinal toxicities.		

Tardive Dyskinesia Needs Assessment

Gap #1: The symptoms of Tardive Dyskinesia (TD) can range widely in severity and impact may be tough to distinguish from other drug-induced movement disorders and its impact on a patient's quality of life is often underestimated.			
Root Causes	**Learning Objectives**		
Recognition of tardive dyskinesia (TD) is complicated by its variable clinical manifestations and significant symptom overlap with other acute and reversible drug-induced movement disorders (e.g., drug-induced parkinsonism), many of which require opposing management strategies.[1] Although TD most commonly presents with mild orofacial movements, severity and impact can vary widely, and 20% to 30% of patients receiving antipsychotic may have two or more drug-induced movement disorders.[2–4] Diagnostic criteria for TD in DSM-5-TR require careful assessment of symptom duration and antipsychotic drug exposure,[5] while TD in ICD-10-CM is poorly defined and thus, frequently miscoded or neglected entirely in clinical and administrative settings.[6–8] These factors contribute to diagnostic uncertainty and inconsistent screening in routine practice, limiting the recognition and tracking of TD.	Distinguish the symptoms of TD from differential diagnoses and understand the impact TD can have on a patient's quality of life.		
Gap #2: Variability in the interpretation and implementation of the Abnormal Involuntary Movement Scale (AIMS)—including uncertainty regarding clinically meaningful change and screening frequency—contributes to inconsistent monitoring and suboptimal treatment for TD.			
Root Causes	**Learning Objectives**		
AIMS is widely accepted as the standard assessment tool for TD in clinical trials and practice;[9] however, limitations complicate its interpretation and use in routine care. Uncertainty regarding the minimal clinically important difference for AIMS score change and a single global score may obscure the distribution and phenomenology of TD symptoms, which can impact prognosis and tolerability.[10–13] In addition, AIMS does not fully capture the impact of TD on quality of life, which may limit comprehensive assessment of a treatment response. These factors may reduce clinician confidence in interpreting clinical data and determining meaningful improvement. Furthermore, variability in institutional policies regarding screening frequency and time constraints in clinical settings contribute to inconsistent implementation of routine AIMS assessments.[10,14] Collectively, these factors create uncertainty in monitoring disease progression and optimizing treatment decision.	Interpret AIMS scores and the nuances of clinically meaningful change, assess the symptom distribution and phenomenology of TD, and implement appropriate screening procedures and monitoring strategies to optimize timely treatment decisions in patients with TD.		
Gap #3: Fragmented care between psychiatry and neurology contributes to delayed recognition, referral, and management of TD, ultimately limiting comprehensive treatment of both psychiatric symptoms and overall disease burden.			
Root Causes	**Learning Objectives**		

TD is an iatrogenic condition associated with the use of dopamine-receptor blocking agents prescribed by psychiatrists, but clinical manifestations of TD vary in severity and impact. Although psychiatrists prioritize stabilization of underlying psychiatric illness, their formal training and clinical emphasis on movement disorder evaluation may vary.[5,15] Conversely, neurologists may have limited early exposure to TD and are frequently consulted only after symptom progression.[16,17] Consequently, limited interdisciplinary communication, inconsistent referral pathways, and insufficient integration of TD monitoring into collaborative care models contribute to the delayed recognition and incomplete management of the full disease burden.	Understand the role each member in a multidisciplinary healthcare team has toward the effective management of symptoms in patients with TD.		

Valbenazine overview.

Approved by the FDA in 2016, valbenazine (Ingrezza) is a vesicular monoamine transporter 2 (VMAT2) inhibitor and is the first medication indicated for the treatment of adults with tardive dyskinesia (TD). It is the purified prodrug of the (+)-alpha-isomer of tetrabenazine, allowing it to be highly selecting in inhibiting VMAT2 with a lower risk of adverse side effects due to off target binding. Valbenazine is available in 40 mg, 60 mg, and 80 mg capsules, is able to be taken with or without food, and has a slower breakdown rate that results in a longer half-life and allows once-daily administration.[18,19] Valbenazine Sprinkle capsules were also developed to address the needs of patients with TD who have difficulties swallowing or do not want to take whole capsules due to motor symptoms, advanced age, or other conditions.[20] Currently, the FDA states possible adverse reactions from valbenazine include somnolence, balance disorders, headache, anticholinergic effects, akathisia, vomiting, nausea, and arthralgia.[21]

The safety and efficacy of valbenazine were established in two published randomized, double-blind, placebo-controlled trials (KINECT-2[18] and KINECT-3[19] which demonstrated valbenazine's effectiveness in improving Abnormal Involuntary Movement (AIMS) score in patients with TD, compared with placebo. Efficacy for valbenazine 60 mg was established using the FDA's model-informed drug development approach and approved for use in 2021.[22] In KINECT-2, patients receiving valbenazine (25–75 mg once daily) demonstrated improvement in AIMS total score at week six versus placebo (least squares mean difference [LSMD] -2.4, P=0.005).[18] KINECT-3 (phase 3 study) patients took either valbenazine 40 mg or 80 mg or placebo. At six weeks, both valbenazine groups saw an improvement in AIMS score from baseline (80 mg: LSMD -3.1, P <0.001; 40 mg LSMD -1.8, P =0.002). Results from J-KINECT (Japanese phase 3 study)[23] were consistent with those from KINECT-3 and its findings supported the approval of valbenazine for TD in Asia. Recently (Dec 2025), Neurocrine Biosciences announced that the KINECT-DCP (phase 3, double-blind, placebo-controlled) clinical trial, which evaluated the use of valbenazine for the treatment of dyskinesia due to cerebral palsy in pediatric and adult populations did not meet its primary or secondary endpoints.[24] Results from this study are forthcoming.

Drug interactions and Competitors

Valbenazine is one of two FDA-approved VMAT2 inhibitors, alongside deutetrabenazine, used for the treatment of TD. Other drugs or drugs that have been used to treat TD in the past include tetrabenazine (another VMAT2 inhibitor used for Huntington's disease), Clonazepam and diazepam (GABA agonists), Gingo biloba (herbal supplement), and Amantadine (NMDA receptor agonist). Clinically significant drug interactions with valbenazine include Monoamine Oxidase Inhibitors (MAOIs), strong CYP2D6 inhibitors, strong CYP3A4 inducers, and digoxin.[20,21]

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